

Yongxin (Kiki) Kong



Chinese Human Reference Atlas Mirror: Setup, Localization, and Usage

B.W. Herr II^{1,*}, L. Liu², R. Eils³, L. Gu⁵, F. Wu², Y. Kong^{1,4}, A. Bueckle¹, K. Börner¹

¹Indiana University, Department of Intelligent Systems Engineering, Bloomington, IN, USA

²Fudan University, Intelligent Medicine Institute, Fudan, Shanghai, China

³Charité, Berlin Institute of Health, Berlin, Germany

⁴Chinese Academy of Sciences, Hong Kong Institute of Science & Innovation, Hong Kong, China

⁵Max Planck Institute for Heart and Lung Research, Bad Nauheim, Germany

bherr@iu.edu

* Corresponding Author

Abstract

The Human Reference Atlas (HRA, <https://humanatlas.io>) is a major international effort across 25+ consortia funded by NIH and CIFAR and with strong support by the Human Cell Atlas (HCA) [<https://www.nature.com/articles/s41586-024-08338-4>] that is constructing a multiscale, multimodal, three-dimensional atlas of the anatomical structures and cells in the healthy human body [<https://www.nature.com/articles/s41592-024-02563-5>]. The HRA details standard terminologies and data structures for describing specimens, biological structures, and spatial positions linked to existing ontologies and also to scholarly publications, called HRAlit [<https://www.nature.com/articles/s41597-024-03416-8>]. The entire HRA Knowledge Graph (KG) is compiled to a set of static files and shared as a compressed archive for mirroring. Mirror sites then host this data from their institution and contribute their own DOs back to the HRA project.

The Chinese Human Reference Atlas Mirror Site (<https://imi.fudan.edu.cn/humanatlas-mirror>) at the Fudan University Medical Science Data Center (FMSDC) in Shanghai aims to provide high-quality, efficient access to HRA data for users in Asia, particularly in China. Key tasks include establishing robust mirror infrastructure for hosting HRA KG data, supporting data contributions from Chinese researchers, ensuring seamless integration with existing biomedical repositories in Asia, facilitating HRA data analysis and integration for biomedical research, providing HRA technical support and training, and promoting HRA resource dissemination within the regional research community.